Join and Lobby Setup Sequence

Player
Client UI (LoginView / LobbyView)
ClientController
ClientSideConnection (SocketClientConnection / RMIClientConnection)
Server transport (SocketServerConnection / RMIServerConnection)
MessageDispatcher
LobbyManager
GameController
LobbyState
ReadyCheckState
StartGameState
VirtualView
ClientModel

Connection setup

alt [Socket]
serialized ClientMessage
[RMI]
remote method invocation
login / reconnect / setReady / setPlayersCount

registerPendingConnection(connection)
LobbyUpdateMessage

Join lobby

insert nickname and choose totem
send(LoginMessage)
send(LoginMessage)

alt [Socket transport]
writeObject(LoginMessage)
[RMI transport]
login(callback, nickname, totemColor)
create LoginMessage

dispatch(connection, LoginMessage)
handleLogin(connection, LoginMessage)

alt [invalid login]
LoginAck(REJECTED, reason)
LoginAckMessage
applyLoginAck(message)
show error
[valid login]
onMessageReceived(nickname, JoinGameAction)
handleAction(ctx, nickname, JoinGameAction)
isLobbyFull()
isNicknameAvailable(nickname)
isTotemAlreadyTaken(totem)

alt [invalid nickname / totem / lobby full]
throws exception
exception
LoginAck(REJECTED, reason)
LoginAckMessage
applyLoginAck(message)
[join accepted]
addLobbyPlayer(new Player)
notifyListeners(onPlayerJoined)

opt [chosenPlayersCount set and lobby full]
changeState(new ReadyCheckState)
onStateChanged(ReadyCheckState)
StateChangedMessage

unregisterPendingConnection(connection)
registerConnection(nickname, connection)
create VirtualView(connection, nickname)
LoginAck(LOGIN_OK)
LobbyUpdateMessage
StateChangedMessage(currentState)
server messages
applyLoginAck()
applyLobbyUpdate()
applyStateChanged()
render lobby

Player count selection

alt [lobby leader selects player count]
choose player count
send(SetPlayersCountMessage)
send(SetPlayersCountMessage)

alt [Socket transport]
writeObject(SetPlayersCountMessage)
[RMI transport]
setPlayersCount(callback, count)
create SetPlayersCountMessage

dispatch(connection, SetPlayersCountMessage)
handleSetPlayersCount(nickname, message)
onMessageReceived(nickname, SetPlayerCountAction)
handleAction(ctx, nickname, SetPlayerCountAction)

alt [sender is not lobby leader]
throws exception
ErrorMessage
applyError(message)
[selected count is invalid]
throws exception
ErrorMessage
applyError(message)
[count accepted and lobby not full]
setChosenPlayersCount(count)
notifyListeners(onPlayersCountSet)
LobbyUpdateMessage
applyLobbyUpdate(message)
[count accepted and lobby full]
setChosenPlayersCount(count)
notifyListeners(onPlayersCountSet)

alt [all players already ready]
changeState(new StartGameState)
[not all players ready]
changeState(new ReadyCheckState)
onStateChanged(ReadyCheckState)
StateChangedMessage
LobbyUpdateMessage

Ready check

loop [each player can toggle ready / not ready]
click Ready / Not ready
send(ReadyMessage)
send(ReadyMessage)

alt [Socket transport]
writeObject(ReadyMessage)
[RMI transport]
setReady(callback, ready)
create ReadyMessage

dispatch(connection, ReadyMessage)
handleReady(nickname, ready)
onMessageReceived(nickname, ReadyAction)

alt [current state is LobbyState]
handleAction(ctx, nickname, ReadyAction)
markPlayerReady / markPlayerNotReady
LobbyUpdateMessage
applyLobbyUpdate(message)
[current state is ReadyCheckState]
handleAction(ctx, nickname, ReadyAction)
markPlayerReady / markPlayerNotReady

alt [all players ready]
changeState(new StartGameState)
[not all players ready]
LobbyUpdateMessage
applyLobbyUpdate(message)

Game start

create tribe deck
create building decks
create Game
game.start()
setGame(game)
notifyListeners(onGameStarted)
changeState(new TotemPlacementState)
onStateChanged(TotemPlacementState)
StateChangedMessage
CurrentPlayerChangedMessage
GameStartedMessage(GameSnapshot)
StateChangedMessage(TotemPlacementState)
GameStartedMessage
applyGameStarted(snapshot)
render game board

Player count changed during ready check

opt [leader changes player count while in ReadyCheckState]
choose new player count
send(SetPlayersCountMessage)
send(SetPlayersCountMessage)
dispatch(connection, SetPlayersCountMessage)
handleSetPlayersCount(nickname, message)
onMessageReceived(nickname, SetPlayerCountAction)
handleAction(ctx, nickname, SetPlayerCountAction)
setChosenPlayersCount(count)
resetReadyPlayers()
notifyListeners(onPlayersCountSet)

alt [lobby no longer full]
changeState(new LobbyState)
onStateChanged(LobbyState)
StateChangedMessage
[lobby still full]
LobbyUpdateMessage

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